

Python Tuple Revision Chart

This revision sheet covers Python tuple properties, mutability, built-in functions, methods, operations, and slicing with start, stop, and step.[cite:22][cite:33][cite:69]

Tuple chart

Section	Number	Topic	Explanation	Example	Output
Properties	1	Syntax	Tuple is written with parentheses ().[cite:22][cite:33]	<code>t = (1, 2, 3)</code>	<code>(1, 2, 3)</code>
Properties	2	Empty tuple	Empty tuple is written as ().[cite:33][cite:27]	<code>t = ()</code>	<code>()</code>
Properties	3	Single element tuple	A single-element tuple needs a trailing comma, otherwise it is just a normal value.[cite:33][cite:27]	<code>t = (5,)</code>	<code>(5,)</code>
Properties	4	Mutability	Tuple is immutable, which means elements cannot be changed after creation.[cite:33][cite:64]	<code>t = (1, 2, 3) t[0] = 10</code>	Error
Properties	5	Ordered	Tuple keeps elements in a fixed order.[cite:22][cite:66]	<code>t = (10, 20, 30)</code>	Order is fixed
Properties	6	Indexing allowed	Tuple supports indexing with positions starting from 0.[cite:22][cite:67]	<code>t = (10, 20, 30)</code> <code>print(t[0])</code>	10
Properties	7	Slicing allowed	Tuple supports slicing with start, stop, and step.[cite:67][cite:69]	<code>t = (0,1,2,3,4,5)</code> <code>print(t[1:5:2])</code>	<code>(1, 3)</code>

Section	Number	Topic	Explanation	Example	Output
Properties	8	Duplicate values	Tuple allows duplicate values.[cite:22] [cite:68]	t = (1, 2, 2, 3)	(1, 2, 2, 3)
Properties	9	Membership testing	Tuple supports <code>in</code> and <code>not in</code> for checking elements. [cite:33][cite:68]	print(2 in (1, 2, 3))	True
Built-in Function	1	len()	Returns total number of elements. [cite:33]	print(len((1, 2, 3)))	3
Built-in Function	2	min()	Returns smallest element. [cite:33]	print(min((3, 1, 2)))	1
Built-in Function	3	max()	Returns largest element. [cite:33]	print(max((3, 1, 2)))	3
Built-in Function	4	sum()	Returns sum of numeric elements. [cite:33]	print(sum((1, 2, 3)))	6
Built-in Function	5	sorted()	Returns a sorted list from the tuple.[cite:33]	print(sorted((3, 1, 2)))	[1, 2, 3]
Built-in Function	6	in	Checks whether an element exists in the tuple.[cite:33]	print(2 in (1, 2, 3))	True
Built-in Function	7	not in	Checks whether an element does not exist in the tuple. [cite:33]	print(5 not in (1, 2, 3))	True
Method	1	count()	Returns how many times a value appears in the tuple. [cite:33]	print((1,2,2,3).count(2))	2
Method	2	index()	Returns the index of the first occurrence of a value.[cite:33]	print((10,20,30).index(20))	1

Section	Number	Topic	Explanation	Example	Output
Operation	1	Concatenation +	Joins two tuples into a new tuple.[cite:67]	<code>print((1,2) + (3,4))</code>	(1, 2, 3, 4)
Operation	2	Repetition *	Repeats the tuple n times. [cite:67]	<code>print((1,2) * 3)</code>	(1, 2, 1, 2, 1, 2)
Operation	3	Membership in	Checks whether an element is present.[cite:33] [cite:68]	<code>print(3 in (1,2,3))</code>	True
Operation	4	Membership not in	Checks whether an element is absent.[cite:33] [cite:68]	<code>print(7 not in (1,2,3))</code>	True

Start, stop, step in slicing

Tuple supports slicing with the syntax `t[start:stop:step]`. [cite:69] In this syntax, `start` is the index where slicing begins, `stop` is the index where slicing ends, and `step` is the gap between selected elements.[cite:69]

Example:

```
t = (0, 1, 2, 3, 4, 5)
print(t[1:5:2])
```

Output:

```
(1, 3)
```

Mutability note

Tuple is immutable, so elements cannot be added, removed, or changed after the tuple is created.[cite:33][cite:64] Because of this, tuples have only a small number of methods compared with mutable types like lists and sets.[cite:33]